Project: Medical Clinic Management System

I want to develop a web application for managing a medical clinic. The goal of this system is to facilitate appointment scheduling, patient data management, medical diagnosis tracking, and provide statistics that help improve clinic management.

1. Authentication & Role Management

The system contains several types of users with different permissions:

- **Patient**
- **Clinic Staff / Receptionist**

- **Doctor**
- **Admin**

A secure login system will be provided for internal users (Doctor, Staff, Admin) with a **Role-Based Access Control (RBAC)** system, where each user has access to their own interfaces and functionalities.

2. Patient Management

For the Patient:

- Create a personal account.
- Update personal information.
- Book a medical appointment based on:
- Available day.

- Available time slot.
- Doctor (if there is more than one doctor).

3. Appointment Management

For Clinic Staff:

They can:

- View all daily and weekly appointments.
- Change appointment status:
- Pending → Confirmed
- Confirmed → Completed
- Organize the appointment schedule.

- Quickly search for an appointment or a patient.

Additional features:

- **Calendar View** to display appointments visually.
- Appointment filtering by:
- Date.
- Doctor.
- Appointment status.

4. Patient Records Management

When a new patient visits the clinic:

- The receptionist creates a medical file

if it does not already exist.

- Enter information such as:
- Full name.
- Date of birth.
- Gender.
- Phone number.
- Address.

Staff members can:

- Search for a patient by name.
- View basic patient information.
- Check previous visit history.

5. Medical Consultation &

Diagnosis System

For the Doctor:

When receiving a patient, the doctor can:

- Open the patient's medical file.
- Write the medical diagnosis.
- Record:
- Medical notes (symptoms).
- Diagnosis.
- Recommended treatment.

Then, the doctor saves the consultation in the patient's medical record.

The doctor can also:

- View all previous diagnoses of the patient.

- Search for any patient by name.

6. Doctor Dashboard

Provides the doctor with various statistics that will be discussed later.

7. Admin Dashboard

The admin has system management permissions.

The admin can:

- Create accounts for doctors and clinic staff.

- View different system statistics.

8. Security

The system must include:

- Authentication using **JWT**.
- Password encryption.
- API protection.
- Role-based authorization.